

Unit I :: Distribution of Velocities and Molecular Collisions

☞ **PHY0500304: Heat & Thermodynamics** ☞

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Syllabus

Unit I: Distribution of Velocities and Molecular Collisions

Maxwell-Boltzmann Law of Distribution of Velocities in an Ideal Gas and its Experimental Verification. Mean, RMS and Most Probable Speeds. Degrees of Freedom. Law of Equipartition of Energy (No proof required). Mean Free Path. Collision Probability. Transport Phenomenon in Ideal Gases: (1) Viscosity, and (2) Thermal Conductivity. Brownian Motion (qualitative idea only).

1 Maxwell-Boltzmann distribution law

The Maxwell-Boltzmann distribution is a result of the kinetic theory of gases, which provides a simplified explanation of many fundamental gaseous properties, including pressure and diffusion. This distribution applies fundamentally to particle velocities in three dimensions, but turns out to depend only on the speed (the magnitude of the velocity) of the particles. A particle speed probability distribution indicates which speeds are more likely: a randomly chosen particle will have a speed selected randomly from the distribution, and is more likely to be within one range of speeds than another.

The kinetic theory of gases applies to the classical ideal gas, which is an idealization of real gases. In real gases, there are various effects (e.g., van der Waals interactions, vortical flow, relativistic speed limits, and quantum exchange interactions) that can make their speed distribution different from the Maxwell–Boltzmann form. However, rarefied gases at ordinary temperatures behave very nearly like an ideal gas and the Maxwell speed distribution is an excellent approximation for such gases. This is also true for ideal plasmas, which are ionized gases of sufficiently low density.

1.1 Postulates of kinetic theory of gases

1. Every mole of a gas contains about 6.023×10^{23} molecules.
2. The gas molecules may be regarded as point masses.
3. The gas molecules are in a state of continuous random motion.
4. In the absence of an external force field, the molecules are distributed uniformly in the container, i.e. an ideal gas behaves as an isotropic medium.
5. The molecules of a gas experience force only during collisions.
6. The molecules of a gas behave as rigid, perfectly elastic hard spheres.
7. The duration of collisions is negligible compared to the time interval between collisions.
8. There is a spread of molecular speeds from zero to infinity.
9. The gravitational potential energy has negligible effect on the motion of the gas molecules.

1.2 Fundamental assumptions

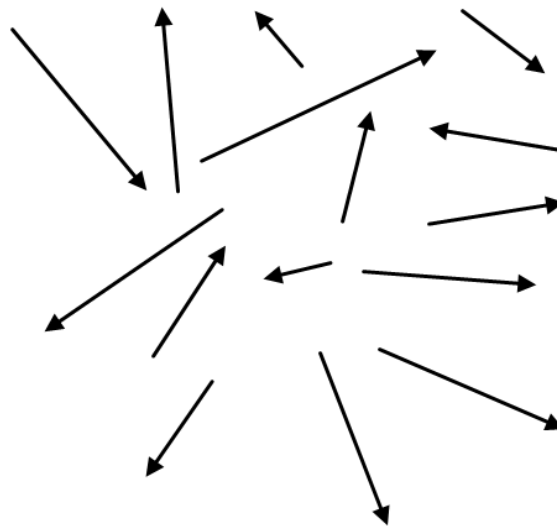
In the derivation of the Maxwell-Boltzmann distribution, we make the following basic assumptions:

1. In the equilibrium state, the molecules have complete randomness of direction and velocity.
2. There is no mass motion or convection current in the body of the gas.

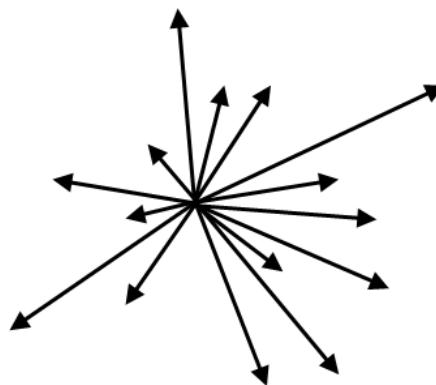
3. The probability of a molecule having an x -velocity component at any point of time is independent of its y - and z -velocity components, and so on.
4. The probability that a molecule selected at random has velocity component in a given range is a function only of the magnitude of the velocity component and the width of the interval.
5. The gas molecules have no rotational or vibrational energies.

1.3 Velocity space

Consider a gas of N molecules enclosed in a vessel, of arbitrary shape, and moving randomly. The velocity vectors are all randomly oriented in all possible directions.



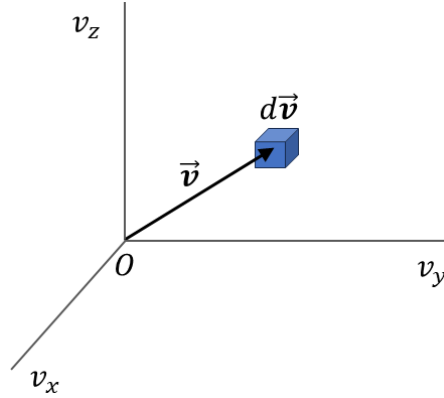
Now we transport each vector parallel to itself to a common origin.



If we now fix the velocity axes v_x , v_y and v_z with respect to a common origin O (that corresponds to particles at rest), every molecular velocity can be represented in this space, which is known as the *velocity space*.

1.4 Derivation of Maxwell-Boltzmann distribution law

Let us now consider an infinitesimal volume element $d\vec{v}$ in this velocity space. The number of velocity vectors that terminate inside $d\vec{v}$ gives us the average number of molecules whose velocities lie between $\vec{v}$ and $\vec{v} + d\vec{v}$.



We have, for any velocity $\vec{v}$, the corresponding speed to be

$$v^2 = v_x^2 + v_y^2 + v_z^2 \quad (1)$$

Let dN_{v_x} be the number of molecules having velocity components in the range v_x and $v_x + dv_x$. Then from assumption #4, we have

$$\begin{aligned} \frac{dN_{v_x}}{N} &= f(v_x) dv_x \\ \text{or, } dN_{v_x} &= N f(v_x) dv_x \end{aligned} \quad (2)$$

In equation (2), $f(v_x)$ is an unknown function that we have to determine.

Also, the probability of finding a molecule with x -velocity component in the range v_x and $v_x + dv_x$ is

$$p_x = \frac{dN_{v_x}}{N} = f(v_x) dv_x \quad (3)$$

Likewise, from assumption #3, we have

$$p_y = f(v_y) dv_y \quad (4)$$

$$p_z = f(v_z) dv_z \quad (5)$$

Then from the law of compound probabilities, the probability of finding a molecule having velocity components simultaneously in the ranges $[v_x, v_x + dv_x]$, $[v_y, v_y + dv_y]$ and $[v_z, v_z + dv_z]$ is

$$\frac{d^3 N_{v_x v_y v_z}}{N} = p_x p_y p_z = f(v_x) f(v_y) f(v_z) dv_x dv_y dv_z \quad (6)$$

Hence, the number density of velocity points is

$$\rho = \frac{d^3 N_{v_x v_y v_z}}{dv_x dv_y dv_z} = N f(v_x) f(v_y) f(v_z) \quad (7)$$

It follows from assumption #1 that the velocity space has been assumed to be isotropic, and so this implies that ρ is independent of the direction of $\vec{v}$ and only depends on v^2 , i.e

$$\begin{aligned}\rho &\propto J(v^2) \\ \Rightarrow f(v_x)f(v_y)f(v_z) &= J(v^2)\end{aligned}\quad (8)$$

where J is some other function.

For a fixed value of v^2 , $J(v^2)$ is constant and so

$$\begin{aligned}d[J(v^2)] &= 0 \\ \Rightarrow d[f(v_x)f(v_y)f(v_z)] &= 0 \\ \Rightarrow \frac{\partial f(v_x)}{\partial v_x} dv_x f(v_y)f(v_z) + f(v_x) \frac{\partial f(v_y)}{\partial v_y} dv_y f(v_z) + f(v_x)f(v_y) \frac{\partial f(v_z)}{\partial v_z} dv_z &= 0\end{aligned}\quad (9)$$

Dividing equation (9) by $f(v_x)f(v_y)f(v_z)$, we obtain

$$\frac{1}{f(v_x)} \frac{\partial f(v_x)}{\partial v_x} dv_x + \frac{1}{f(v_y)} \frac{\partial f(v_y)}{\partial v_y} dv_y + \frac{1}{f(v_z)} \frac{\partial f(v_z)}{\partial v_z} dv_z = 0 \quad (10)$$

Also, for a fixed value of v^2 , we have

$$\begin{aligned}d[v^2] &= 0 \\ \Rightarrow d[v_x^2 + v_y^2 + v_z^2] &= 0 \\ \Rightarrow v_x dv_x + v_y dv_y + v_z dv_z &= 0\end{aligned}\quad (11)$$

Suppose that B is some unknown constant. Then multiplying equation (11) with $2B$ and adding to equation (10) gives

$$\left[\frac{1}{f(v_x)} \frac{\partial f(v_x)}{\partial v_x} + 2B v_x \right] dv_x + \left[\frac{1}{f(v_y)} \frac{\partial f(v_y)}{\partial v_y} + 2B v_y \right] dv_y + \left[\frac{1}{f(v_z)} \frac{\partial f(v_z)}{\partial v_z} + 2B v_z \right] dv_z = 0 \quad (12)$$

Even though v is held constant, v_x , v_y , v_z may individually change and yet satisfy equation (1). In equation (12), the first term within brackets depends only on v_x , the second term on v_y and the third term on v_z .

Because dv_x , dv_y , dv_z are arbitrary, the only way for equation (12) to hold good is if all three terms vanish identically.

Therefore, for the v_x term, we would have

$$\begin{aligned}\frac{1}{f(v_x)} \frac{\partial f(v_x)}{\partial v_x} + 2B v_x &= 0 \\ \Rightarrow \frac{\partial f(v_x)}{\partial v_x} &= -2B v_x f(v_x) \\ \Rightarrow \frac{\partial f(v_x)}{f(v_x)} &= -2B v_x dv_x \\ \Rightarrow \int \frac{df}{f} &= -2B \int v_x dv_x \\ \Rightarrow \ln f(v_x) &= -2B \frac{v_x^2}{2} + \ln A\end{aligned}$$

where $\ln A$ is the constant of integration. From the above we obtain

$$\begin{aligned}
 \ln f(v_x) - \ln A &= -B v_x^2 \\
 \Rightarrow \ln \frac{f(v_x)}{A} &= -B v_x^2 \\
 \Rightarrow \frac{f(v_x)}{A} &= e^{-B v_x^2} \\
 \Rightarrow f(v_x) &= A e^{-B v_x^2}
 \end{aligned} \tag{13}$$

Likewise, from the v_y and v_z terms, we obtain

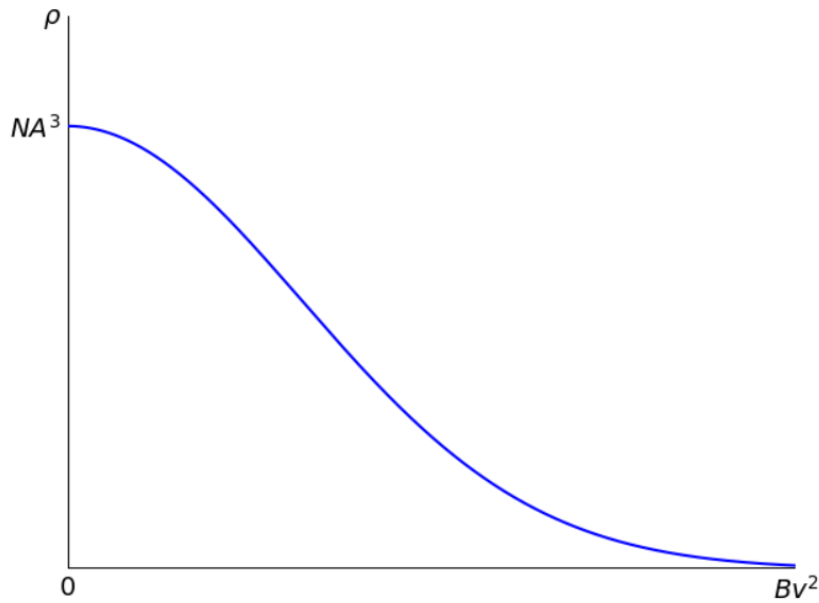
$$f(v_y) = A e^{-B v_y^2} \tag{14}$$

$$f(v_z) = A e^{-B v_z^2} \tag{15}$$

Using equations (13), (14) and (15) in equation (7) gives us

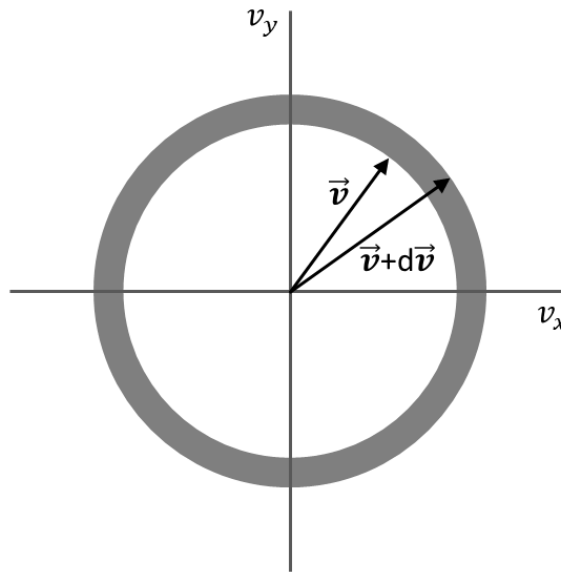
$$\begin{aligned}
 \rho &= N \left(A e^{-B v_x^2} \right) \left(A e^{-B v_y^2} \right) \left(A e^{-B v_z^2} \right) \\
 &= N A^3 e^{-B(v_x^2 + v_y^2 + v_z^2)} \\
 \Rightarrow \boxed{\rho(v) = N A^3 e^{-B v^2}}
 \end{aligned} \tag{16}$$

Equation (16) is known as *Maxwell's velocity density distribution function*, where ρ varies with molecular speeds v as shown in the plot below:



1.4.1 Molecular distribution of speeds

Consider a spherical shell of radius v and thickness dv in the velocity space. Then all points within this shell correspond to molecules with speeds in the range v and $v + dv$. In the spherical



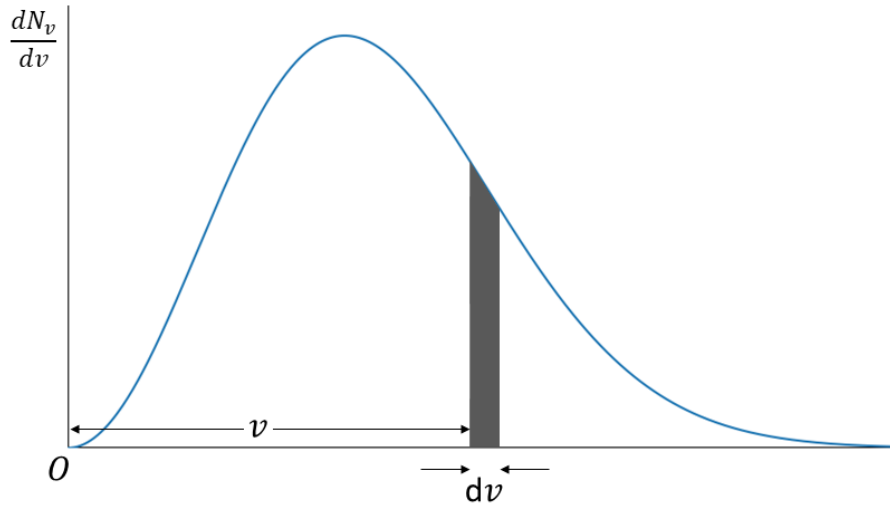
polar coordinates, the volume of an arbitrary volume element is

$$dv_x dv_y dv_z = v^2 \sin \theta d\theta d\phi dv \quad (1)$$

The number of molecules with speeds in the range v and $v + dv$ is obtained by integrating ρ over this shell as

$$\begin{aligned} dN_v &= d^3 N_{v_x v_y v_z} \\ &= \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} N A^3 e^{-Bv^2} v^2 \sin \theta d\theta d\phi dv \\ &= N A^3 e^{-Bv^2} v^2 dv \int_0^{\pi} \sin \theta d\theta \int_0^{2\pi} d\phi \\ &= N A^3 e^{-Bv^2} v^2 dv [\cos 0 - \cos \pi] 2\pi \\ &= N A^3 e^{-Bv^2} v^2 dv \times 2 \times 2\pi \\ \Rightarrow dN_v &= 4\pi N A^3 v^2 e^{-Bv^2} dv \\ \Rightarrow \boxed{\frac{dN_v}{dv} &= 4\pi N A^3 v^2 e^{-Bv^2}} \quad (2) \end{aligned}$$

Equation (2) gives us the *Maxwellian distribution function of molecular speeds*.



In the plot of $\frac{dN_v}{dv}$ with respect to v , the area of a vertical shaded strip under the graph gives the number of molecules which have speeds between v and $v + dv$. Hence, the total area under the entire curve gives us the total number of molecules N .

1.4.2 Determination of the constants

The total number of molecules N can be obtained by integrating dN_v over all speeds, i.e.

$$N = \int dN_v = 4\pi N A^3 \int_0^{\infty} v^2 e^{-Bv^2} dv \quad (1)$$

We now use the following standard integral:

$$\int_0^{\infty} e^{-ax^2} x^n dx = \frac{1}{2a^{(n+1)/2}} \Gamma\left(\frac{n+1}{2}\right) \quad (2)$$

where the gamma function is defined as

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$$

Some important values of $\Gamma(x)$ are given below:

x	$\Gamma(x)$
1/2	$\sqrt{\pi}$
1	1
3/2	$\sqrt{\pi}/2$
2	1

The gamma function also satisfies the recurrence relation: $\Gamma(n+1) = n\Gamma(n)$

In equation (2), setting $x \rightarrow v$, $\alpha \rightarrow B$ and $n = 2$ gives us

$$\begin{aligned}
 \int_0^{\infty} e^{-Bv^2} v^2 dv &= \frac{1}{2B^{(2+1)/2}} \Gamma\left(\frac{2+1}{2}\right) \\
 &= \frac{1}{2B^{3/2}} \Gamma\left(\frac{3}{2}\right) \\
 \Rightarrow \int_0^{\infty} v^2 e^{-Bv^2} dv &= \frac{\sqrt{\pi}}{4B^{3/2}}
 \end{aligned} \tag{3}$$

Inserting equation (3) in equation (1), we get

$$\begin{aligned}
 \mathcal{N} &= 4\pi N A^3 \times \frac{\sqrt{\pi}}{4B^{3/2}} \\
 \Rightarrow B^{3/2} &= \sqrt{\pi^3} A^3 \\
 \Rightarrow A^3 &= \frac{B^{3/2}}{\sqrt{\pi^3}} = \frac{B^{3/2}}{\pi^{3/2}} = \left(\sqrt{\frac{B}{\pi}}\right)^3 \\
 \Rightarrow A &= \sqrt{\frac{B}{\pi}}
 \end{aligned} \tag{4}$$

However, equation (4) only gives us the constant A in terms of another constant B . We can find B using the law of equipartition of energy of a gas.

The average value of v^2 is obtained as

$$\langle v^2 \rangle = \frac{\int_0^{\infty} v^2 dN_v}{\int_0^{\infty} dN_v} \tag{5}$$

Now, we have

$$\begin{aligned}
 \int_0^{\infty} v^2 dN_v &= 4\pi N A^3 \int_0^{\infty} v^4 e^{-Bv^2} dv \\
 &= 4\pi N A^3 \frac{\Gamma(5/2)}{2B^{5/2}} \quad [\text{using the standard integral}] \\
 \Rightarrow \int_0^{\infty} v^2 dN_v &= \frac{2\pi N A^3}{B^{5/2}} \Gamma\left(\frac{5}{2}\right)
 \end{aligned}$$

Also, we have

$$\int_0^{\infty} dN_v = 4\pi N A^3 \int_0^{\infty} v^2 e^{-Bv^2} dv = \frac{2\pi N A^3}{B^{3/2}} \Gamma\left(\frac{3}{2}\right)$$

Using the above two integrals in equation (5), we obtain

$$\begin{aligned}
 \langle v^2 \rangle &= \frac{2\pi N A^3 \Gamma(5/2)}{B^{5/2}} \cdot \frac{B^{3/2}}{2\pi N A^3 \Gamma(3/2)} \\
 &= \frac{B^{3/2}}{B^{5/2}} \cdot \frac{\Gamma(5/2)}{\Gamma(3/2)} \\
 &= \frac{B^{3/2}}{B^{5/2}} \cdot \frac{3/2 \Gamma(3/2)}{\Gamma(3/2)} \\
 &= \frac{3}{2} B^{3/2-5/2} = \frac{3}{2} B^{-1} \\
 \Rightarrow \langle v^2 \rangle &= \frac{3}{2B}
 \end{aligned} \tag{6}$$

Hence, the average kinetic energy of the gas molecules becomes

$$\begin{aligned}
 \langle K \rangle &= \frac{1}{2} m \langle v^2 \rangle = \frac{1}{2} m \times \frac{3}{2B} \\
 \Rightarrow \langle K \rangle &= \frac{3m}{4B}
 \end{aligned} \tag{7}$$

However, from the law of equipartition of energy of a gas, the average kinetic energy of the gas molecules at temperature T is

$$\langle K \rangle = \frac{3}{2} k_B T \tag{8}$$

where $k_B = 1.381 \times 10^{-23} \text{ J K}^{-1}$ is the Boltzmann constant.

Comparing equations (7) and (8) gives us

$$\begin{aligned}
 \frac{3m}{4B} &= \frac{3}{2} k_B T \\
 \Rightarrow \frac{3m}{4B} &= \frac{3}{2} k_B T \\
 \Rightarrow B &= \frac{m}{2k_B T}
 \end{aligned} \tag{9}$$

Inserting equation (9) in equation (4) gives us the value of the constant A :

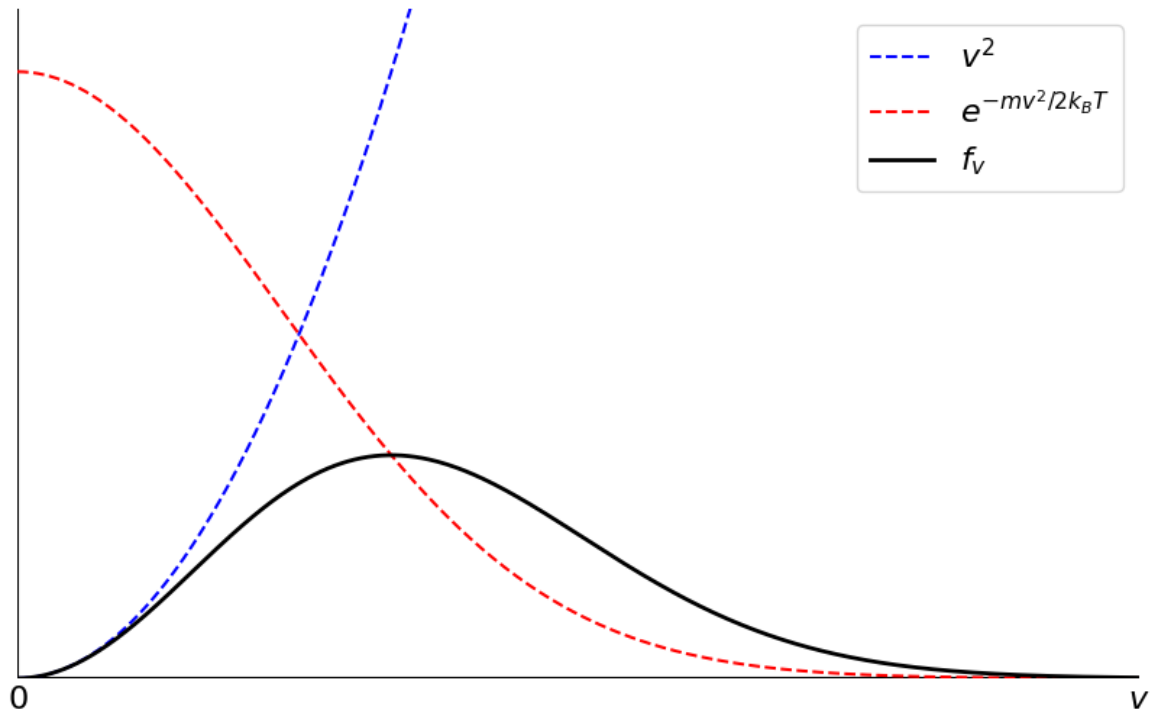
$$\begin{aligned}
 A &= \sqrt{\frac{B}{\pi}} \\
 \Rightarrow A &= \sqrt{\frac{m}{2\pi k_B T}}
 \end{aligned} \tag{10}$$

Finally, substituting for the constants A and B using equations (10) and (9) respectively in the Maxwellian distribution function for molecular speeds, we get

$$\begin{aligned}
 dN_v &= 4\pi N \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T} dv \\
 \Rightarrow f_v &= \frac{dN_v}{N dv} = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}
 \end{aligned} \tag{11}$$

The *Maxwellian density function* f_v , from equation (11), gives us the probability per unit molecular speed of finding a molecule with speed between v and $v + dv$. This function has the following features:

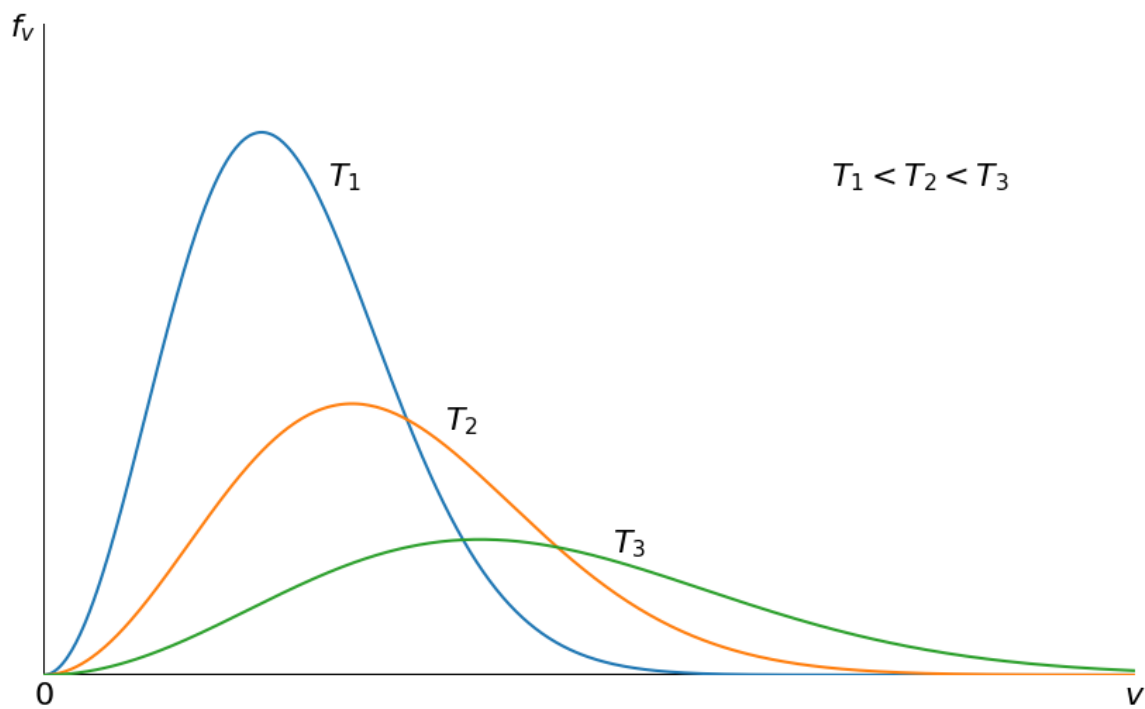
1. $f_v = 0$ for $v = 0$ (due to v^2) and $f_v \rightarrow 0$ for $v \rightarrow \infty$ (due to $e^{-mv^2/2k_B T}$).
2. For small molecular speeds, the v^2 factor dominates and f_v increases quadratically.
3. As the speed increases, the $e^{-mv^2/2k_B T}$ factor begins to dominate and the distribution starts decaying exponentially after attaining a maximum value.



4. Because f_v is a probability density function, at all T the following condition must hold good:

$$\int_0^{\infty} f_v dv = 1$$

5. As the temperature T increases, more and more molecules gain kinetic energy and so the peak shifts towards higher molecular speeds. But the total number of molecules remains constant, and so to keep the corresponding area under the curve constant, curve becomes progressively flatter with increasing T .



Exercises

1. Calculate the probability that the speed of an O_2 molecule will lie between 100 m/s and 101 m/s at 200 K. The mass of an O_2 molecule is 32 u. Take $k_B = 1.381 \times 10^{-23} \text{ J K}^{-1}$ and $N_A = 6.023 \times 10^{26} \text{ kmol}^{-1}$.

2 Molecular speeds

2.1 Average or mean speed

The average speed of the gas is defined as

$$v_{\text{avg}} = \langle v \rangle = \frac{1}{N} \int_0^{\infty} v \, dN_v$$

Substituting for dN_v from the Maxwellian density function, we have

$$\begin{aligned} v_{\text{avg}} &= \frac{1}{N} \times 4\pi N \left(\frac{m}{2\pi k_B T} \right)^{3/2} \underbrace{\int_0^{\infty} v^3 e^{-mv^2/2k_B T} dv}_1 \\ &= \frac{1}{N} \times 4\pi N \left(\frac{m}{2\pi k_B T} \right)^{3/2} \frac{1}{2(m/2k_B T)^2} \\ &= 2\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \left(\frac{2k_B T}{m} \right)^2 \\ &= \frac{2\pi^{1-3/2} m^{3/2-2}}{2^{3/2-2} k_B^{3/2-2} T^{3/2-2}} = \frac{2^{3/2} k_B^{1/2} T^{1/2}}{\pi^{1/2} m^{1/2}} \\ &= \left(\frac{2^3 k_B T}{\pi m} \right)^{1/2} \\ \Rightarrow v_{\text{avg}} &= \sqrt{\frac{8k_B T}{\pi m}} \end{aligned}$$

2.2 Root mean squared (RMS) speed

The root mean squared (RMS) speed is defined as

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle}$$

Now, we already have that

$$\langle v^2 \rangle = \frac{3}{2B} \quad \text{and} \quad B = \frac{m}{2k_B T}$$

The above give us

$$\langle v^2 \rangle = \frac{3}{2} \times \frac{2k_B T}{m} = \frac{3k_B T}{m}$$

Therefore the RMS speed is

$$v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$$

2.3 Most probable speed

The most probable speed v_p is defined as the speed at which the probability of finding a molecule is maximum.

- This is obtained by finding the root of the equation $\frac{df_v}{dv} = 0$.

We have

$$f_v = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T} \quad (1)$$

Differentiating equation (1) with respect to v and equating to 0 gives $v = v_p$, we have

$$\begin{aligned} \frac{df_v}{dv} &= 0 \\ \Rightarrow 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \frac{d}{dv} [v^2 e^{-mv^2/2k_B T}] &= 0 \\ \Rightarrow \frac{d}{dv} [v^2 e^{-mv^2/2k_B T}] &= 0 \\ \Rightarrow 2v_p e^{-mv^2/2k_B T} + v_p^2 \left(-\frac{m}{k_B T} \right) 2v_p e^{-mv^2/2k_B T} &= 0 \\ \Rightarrow 2v_p e^{-mv^2/2k_B T} \left[1 - \frac{mv_p^2}{k_B T} \right] &= 0 \end{aligned}$$

On the LHS of the above, the factor $e^{-mv^2/2k_B T}$ cannot be zero for any finite value of v . Hence we must have

$$\begin{aligned} 1 - \frac{mv_p^2}{k_B T} &= 0 \\ \Rightarrow \frac{mv_p^2}{k_B T} &= 1 \\ \Rightarrow v_p^2 &= \frac{k_B T}{m} \\ \Rightarrow v_p &= \sqrt{\frac{k_B T}{m}} \end{aligned}$$

Summary

The three molecular speeds are as follows:

$$v_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}, \quad v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}, \quad v_p = \sqrt{\frac{k_B T}{m}}$$

This gives their ratios to be

$$v_{\text{avg}} : v_{\text{rms}} : v_p = \sqrt{\frac{8}{\pi}} : \sqrt{3} : \sqrt{2} \approx 1.6 : 1.7 : 1.4$$

The above shows that

$$v_p < v_{\text{avg}} < v_{\text{rms}}$$

